

✓ Question 1 – OOP & Inheritance (10 marks)

Create a new class `Receptionist` that extends `HotelStaff`.

This class should have:

- A field `languageSpoken` (String)
- A constructor that initializes name, surname, and language
- `toString()` method that shows all information including staff ID and DOB

```
public class Receptionist extends HotelStaff {
    private String languageSpoken;

    public Receptionist(String name, String surname, String language) {
        super(name, surname);
        this.languageSpoken = language;
    }

    public String getLanguageSpoken() {
        return languageSpoken;
    }

    @Override
    public String toString() {
        return super.toString() + " Receptionist - Language: " + languageSpoken;
    }
}
```

✓ Question 2 – Interfaces & Polymorphism (10 marks)

Update the interface `HotelManager` to include the following new method:

```
void listHousekeepersByExperience(int minimumYears);
```

Then, in `WestminsterHotelManager`, implement this method to print out only those housekeepers with years of experience greater than or equal to `minimumYears`.

```
@Override
public void listHousekeepersByExperience(int minimumYears) {
    for (HotelStaff s : hotelStaffList) {
        if (s instanceof HouseKeeper) {
```

```

        if (((HouseKeeper) s).getYearsOfExperience() >= minimumYears) {
            System.out.println(s);
        }
    }
}

```

✓ Question 3 – Collections & Comparator (10 marks)

Create a **Comparator** class named **SortGuestByRoomNumber** that sorts guests based on room number in **ascending order**.

Then write a method:

```
void sortAndPrintGuestsByRoom();
```

that uses this comparator and prints guest info after sorting.

```

public class SortGuestByRoomNumber implements Comparator<Guest> {
    public int compare(Guest a, Guest b) {
        return Integer.compare(a.getRoomNumber(), b.getRoomNumber());
    }
}

public void sortAndPrintGuestsByRoom() {
    Collections.sort(hotelGuestList, new SortGuestByRoomNumber());
    for (Guest g : hotelGuestList) {
        System.out.println(g);
    }
}

```

✓ Question 4 – Exception Handling (10 marks)

Create a custom exception class **EmptyGuestNameException**.

Throw this exception if someone tries to add a guest with an empty name or surname.

Update the **addGuest()** method to include this check using:

```

if (name.trim().isEmpty() || surname.trim().isEmpty()) {
    throw new EmptyGuestNameException("Name or surname cannot be empty.");
}

public class EmptyGuestNameException extends Exception {

```

```

    public EmptyGuestNameException(String msg) {
        super(msg);
    }
}

// In addGuest() method:
if (name.trim().isEmpty() || surname.trim().isEmpty()) {
    throw new EmptyGuestNameException("Name or surname cannot be empty.");
}

```

✓ Question 5 – Multithreading (10 marks)

Create a thread that prints each staff member's full name (name + surname) with a **1-second delay** between each.

Then write code to **start and run** this thread from `runMenu()` when a user presses option 5.

```

public class StaffPrinterThread extends Thread {
    private ArrayList<HotelStaff> staffList;

    public StaffPrinterThread(ArrayList<HotelStaff> staffList) {
        this.staffList = staffList;
    }

    public void run() {
        for (HotelStaff s : staffList) {
            System.out.println(s.getName() + " " + s.getSurname());
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}

```

✓ Question 6 – GUI Enhancement (10 marks)

In `HotelTableGUI`, add a **label at the bottom** of the window that shows:

"Total Staff Members: X"

Where X is the number of staff in the list. It should update whenever the GUI is opened.

✓ Question 6 – GUI enhancement

```
JLabel countLabel = new JLabel("Total Staff Members: " + list.size());
add(countLabel, BorderLayout.SOUTH);
```

✓ Question 7 – File Handling (10 marks)

Write a method:

```
void saveGuestsToTextFile(String fileName);
```

This should write each guest's name, surname, room number, and nights stayed to the file in this format:

John Doe – Room: 101, Nights: 3

```
public void saveGuestsToTextFile(String fileName) {
    try (PrintWriter writer = new PrintWriter(new FileWriter(fileName))) {
        for (Guest g : hotelGuestList) {
            writer.println(g.getName() + " " + g.getSurname() + " – Room: " +
                g.getRoomNumber() + ", Nights: " + g.getNightsStayed());
        }
    } catch (IOException e) {
        System.out.println("File write error: " + e.getMessage());
    }
}
```

✓ Question 8 – Advanced OOP + Logic (10 marks)

Add a method:

```
public List<HotelStaff> getStaffByAgeRange(int minAge, int maxAge)
```

This should return all staff whose age falls within the given range (inclusive). Assume `getDateOfBirth()` is already available

```
public List<HotelStaff> getStaffByAgeRange(int minAge, int maxAge) {
    List<HotelStaff> result = new ArrayList<>();
    for (HotelStaff s : hotelStaffList) {
        long age = ChronoUnit.YEARS.between(s.getDateOfBirth(),
            LocalDate.now());
        if (age >= minAge && age <= maxAge) {
            result.add(s);
        }
    }
}
```

```
    return result;
}
```

✓ Question 9 – Mixed Logic + File + GUI (Bonus / Optional)

Write a method that:

1. Reads all guests from a text file (format: **Name, Surname, Room, Nights**)
2. Adds them to the system
3. Then displays the full guest list using a GUI window

```
public void importGuestsFromFileAndShowGUI(String fileName) {
    try (Scanner sc = new Scanner(new File(fileName))) {
        while (sc.hasNextLine()) {
            String[] parts = sc.nextLine().split(",");
            if (parts.length == 4) {
                Guest g = new Guest(parts[0].trim(), parts[1].trim());
                g.setRoomNumber(Integer.parseInt(parts[2].trim()));
                g.setNightsStayed(Integer.parseInt(parts[3].trim()));
                hotelGuestList.add(g);
            }
        }
    } catch (IOException e) {
        System.out.println("Read error: " + e.getMessage());
    }
    GuestTableGUI gui = new GuestTableGUI(hotelGuestList);
    gui.setVisible(true);
}
```